SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	CSE	**Date:**	9/10/2026

Code of Conduct

I U.VINAY KUMAR (192311348) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: U. Vinaykumar

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

- Develop a concept map that evaluates resource management strategies in cellular networks. Relate channel allocation, power control, interference management, frequency reuse, and handoff management. Assess which strategy contributes most

to efficient network utilization. Support your evaluation through concept relationships.

Concept map points: Channel Allocation, Power Control, Interference Management, Frequency Reuse and Handoff Management

- Resource management → *is achieved through* → channel allocation, power control, interference management, frequency reuse and handoff management for efficient use of spectrum, power and network capacity.
- Channel allocation → *assigns* → available channels to users and cells and avoids channel conflicts, which improves capacity.
- Channel allocation → *works with* → power control and interference management to assign channels with less interference.
- Power control → *adjusts* → transmission power to the required level, reducing unnecessary power usage and saving energy.
- Power control → *supports* → interference management and frequency reuse by reducing interference between users and cells.
- Interference management → *reduces* → unwanted signals from other users and cells, which improves signal quality.
- Interference management → *coordinates with* → channel allocation and power control.
- Interference management → *enables* → frequency reuse, since reused frequencies are usable only when interference is controlled.
- Frequency reuse → *reuses* → the same frequencies in sufficiently separated cells, which increases capacity.
- Frequency reuse → *requires* → interference control and handoff management, because users keep crossing between closely spaced cells.
- Handoff management → *transfers* → active calls or data to another cell as the user moves, and needs available channels in neighboring cells.
- Handoff management → *maintains* → connection continuity and reduces dropped calls.
- All five strategies → *together produce* → higher spectrum efficiency, better coverage, fewer dropped calls and improved quality of service.

Evaluation: Frequency reuse, supported by interference management, contributes most directly to overall network capacity and spectrum utilization. The reasons are:

- Spectrum is limited in cellular networks.
- Frequency reuse allows the same frequencies to serve multiple cells, which increases the number of users served simultaneously.
- Interference management and power control reduce the harmful effects of reusing frequencies.

- Channel allocation distributes resources effectively, while handoff management maintains connections as users move.

Conclusion: All five strategies are interdependent, so the best network utilization is achieved by combining all of them rather than relying on only one.

